Reading Notes: Analytic Combinatorics

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Chapter 1

COMBINATORIAL STRUCTURES AND ORDINARY GENERATING FUNCTIONS

1.1 Symbolic enumeration methods

▷ **I.2.** *Unimodal permutations.* A unimodal permutation can be viewed as a sequence of elements increasing to a peak and then decreasing. This structure is represented by:

$$\mathcal{P}^{\mathcal{U}} = \textsc{Set}(\mathcal{Z}) \star \mathcal{Z}^{\blacksquare} \star \textsc{Set}(\mathcal{Z})$$

$$P^U(z) = \int_0^z (e^t \cdot \frac{d}{dt} t \cdot e^t) dt = \frac{1}{2}(e^{2z} - 1)$$

which yields

$$P_n^U = n![z^n]\frac{1}{2}e^{2z} = n!\frac{1}{2}\frac{2^n}{n!} = 2^{n-1}$$

Alternatively,

$$P_n^U = \sum_{k=1}^{n} \binom{n-1}{k-1} = \sum_{j=0}^{n-1} \binom{n-1}{j} = 2^{n-1}$$

where k is the position of n in the permutation.

1.2 Admissible constructions and specifications

1.3 Integer compositions and partitions

▷ **I.16.** *Partitions with summands bounded in number and size.* The number of partitions of size n with at most k summands, each at most l, is

$$[z^n]\frac{(1-z)(1-z^2)\cdots(1-z^{k+l})}{((1-z)(1-z^2)\cdots(1-z^k))\cdot((1-z)(1-z^2)\cdots(1-z^l))} \tag{1.1}$$

Let $f(k,l)$ denote the generating function where the coefficient of z^n represents the number of such partitions. To derive a functional equation, consider the value of the largest

part. There are two mutually exclusive possibilities: the largest part is strictly less than l, or it is exactly l. This distinction yields:

$$f(k,l) = f(k,l-1) + z^l f(k-1,l) \tag{1.2}$$

We can derive 1.1 from this relation by introducing $F_k(x)$, the generating function for $f(k,l)$ with respect to the index l:

$$F_k(x) = \sum_{l=0}^{\infty} f(k,l)x^l$$

For the base case $k = 0$, the only valid partition is the empty partition of size 0, implying $f(0,l) = 1$ for all $l \geq 0$. Thus:

$$F_0(x) = \sum_{l=0}^{\infty} 1 \cdot x^l = \frac{1}{1-x}$$

Summing 1.2 over $l \geq 1$ after multiplying by x^l, we obtain:

$$\sum_{l=1}^{\infty} f(k,l)x^l - \sum_{l=1}^{\infty} f(k,l-1)x^l = \sum_{l=1}^{\infty} f(k-1,l)(zx)^l$$

- The first term simplifies to $F_k(x) - 1$, since $f(k,0) = 1$.
- The second term is simply $xF_k(x)$.
- The right-hand side yields $F_{k-1}(zx) - 1$.

Plugging these in:

$$(1-x)F_k(x) - 1 = F_{k-1}(zx) - 1 \implies F_k(x) = \frac{1}{1-x}F_{k-1}(zx)$$

Now we apply this functional relation repeatedly:

$$\begin{aligned} F_k(x) &= \frac{1}{1-x}F_{k-1}(zx) \\ &= \frac{1}{1-x}\frac{1}{1-zx}F_{k-2}(z^2x) \\ &= \frac{1}{(1-x)(1-zx)\cdots(1-z^{k-1}x)}F_0(z^kx) \end{aligned}$$

Since $F_0(x) = \frac{1}{1-x}$, we have $F_0(z^kx) = \frac{1}{1-z^kx}$. Thus:

$$F_k(x) = \sum_{l=0}^{\infty} f(k,l)x^l = \frac{1}{(1-x)(1-zx)\cdots(1-z^kx)}$$

Since all the roots of the denominator are distinct (assuming $|z| < 1$ or that z is a formal variable), we can decompose this into a sum of partial fractions:

$$\frac{1}{\prod_{i=0}^{k}(1-z^ix)} = \sum_{j=0}^{k} \frac{A_j}{1-z^jx}$$

To find a specific A_j, we multiply both sides by $(1 - z^j x)$ and then set $x = z^{-j}$:

$$\begin{aligned} A_j &= \left[\frac{1 - z^j x}{\prod_{i=0}^{k}(1 - z^i x)}\right]_{x=z^{-j}} \\ &= \frac{1}{\prod_{i=0}^{j-1}(1 - z^i z^{-j}) \prod_{i=j+1}^{k}(1 - z^i z^{-j})} \\ &= \frac{(-1)^j z^{j(j+1)/2}}{\prod_{i=1}^{j}(1 - z^i) \prod_{i=1}^{k-j}(1 - z^i)} \end{aligned}$$

The coefficient of x^l in $F_k(x)$ is the sum of the coefficients of x^l in each partial fraction. For a term $\frac{A_j}{1-z^j x}$, the coefficient is $A_j (z^j)^l$.

$$\begin{aligned} f(k,l) &= \sum_{j=0}^{k} A_j (z^j)^l \\ &= \sum_{j=0}^{k} \frac{(-1)^j z^{j(j+1)/2} \cdot z^{jl}}{\prod_{i=1}^{j}(1 - z^i) \prod_{i=1}^{k-j}(1 - z^i)} \\ &= \frac{1}{\prod_{i=1}^{k}(1 - z^i)} \sum_{j=0}^{k} \binom{k}{j}_z (-1)^j z^{j(j+1)/2 + jl} \end{aligned}$$

where $\binom{k}{j}_z$ is the Gaussian binomial coefficient:

$$\begin{aligned} \binom{k}{j}_z &= \frac{\prod_{i=1}^{k}(1 - z^i)}{\prod_{i=1}^{j}(1 - z^i) \prod_{i=1}^{k-j}(1 - z^i)} \\ &= \frac{(1 - z^k)(1 - z^{k-1}) \cdots (1 - z^{k-j+1})}{(1 - z)(1 - z^2) \cdots (1 - z^j)} \end{aligned}$$

We can rewrite the exponent in the sum:

$$\frac{j(j+1)}{2} + jl = \frac{j(j-1)}{2} + j(l+1)$$

Now the sum becomes

$$f(k,l) = \frac{1}{\prod_{i=1}^{k}(1 - z^i)} \sum_{j=0}^{k} \binom{k}{j}_z (-1)^j z^{j(j-1)/2} (z^{l+1})^j$$

The q-binomial theorem states that for a variable X:

$$\prod_{i=0}^{k-1}(1 - z^i X) = \sum_{j=0}^{k} \binom{k}{j}_z (-1)^j z^{j(j-1)/2} X^j \tag{1.3}$$

Plugging $X = z^{l+1}$ into the theorem yields 1.1:

$$\begin{aligned} f(k,l) &= \frac{(1 - z^{l+1})(1 - z^{l+2}) \cdots (1 - z^{l+k})}{(1 - z^1)(1 - z^2) \cdots (1 - z^k)} \\ &= \frac{(1 - z)(1 - z^2) \cdots (1 - z^{k+l})}{((1 - z)(1 - z^2) \cdots (1 - z^k)) \cdot ((1 - z)(1 - z^2) \cdots (1 - z^l))} \end{aligned}$$

To prove 1.3, let $P_k(X)$ be the product we are expanding:

$$P_k(X) = \prod_{i=0}^{k-1}(1 - z^i X)$$

We want to show that $P_k(X) = \sum_{j=0}^{k} \binom{k}{j}_z (-1)^j z^{j(j-1)/2} X^j$. Consider the relationship between $P_k(X)$ and $P_{k-1}(X)$. By definition:

$$P_k(X) = (1 - z^{k-1}X)P_{k-1}(X)$$

Now notice what happens if we replace X with zX in $P_k(X)$:

$$P_k(zX) = \prod_{i=0}^{k-1}(1 - z^{i+1}X) = \frac{1 - z^k X}{1 - X} P_k(X)$$

Rearranging this gives us a functional equation for $P_k(X)$:

$$(1 - X)P_k(zX) = (1 - z^k X)P_k(X)$$

Let $P_k(X) = \sum_{j=0}^{k} c_j X^j$. We substitute this power series into the functional equation:

$$(1 - X)\sum_{j=0}^{k} c_j (zX)^j = (1 - z^k X)\sum_{j=0}^{k} c_j X^j$$

Now, we compare the coefficients of X^j on both sides. For a specific power j:

$$c_j z^j - c_{j-1} z^{j-1} = c_j - c_{j-1} z^k$$

Rearranging to solve for c_j in terms of c_{j-1}:

$$c_j = c_{j-1}\frac{z^{j-1} - z^k}{z^j - 1} = -c_{j-1} z^{j-1}\frac{1 - z^{k-j+1}}{1 - z^j}$$

Starting from the base case $c_0 = P_k(0) = 1$, we can find c_j by iteration:

$$c_1 = -c_0 z^0 \frac{1 - z^k}{1 - z^1}, \quad c_2 = -c_1 z^1 \frac{1 - z^{k-1}}{1 - z^2} = (-1)^2 z^{0+1}\frac{(1 - z^k)(1 - z^{k-1})}{(1 - z^1)(1 - z^2)}$$

The exponent of z becomes $0 + 1 + 2 + \cdots + (j-1) = \frac{j(j-1)}{2}$. The fraction is the definition of the Gaussian binomial coefficient $\binom{k}{j}_z$. Thus, the coefficient is:

$$c_j = (-1)^j z^{j(j-1)/2}\binom{k}{j}_z$$

Plugging the coefficients back into the sum, we obtain:

$$\prod_{i=0}^{k-1}(1 - z^i X) = \sum_{j=0}^{k}\binom{k}{j}_z (-1)^j z^{j(j-1)/2} X^j$$

This completes the proof.

▷ **I.17.** *Systems of linear Diophantine inequalities.*

$$\{x_1 \geq 0,\ x_2 \geq 2x_1,\ x_3 \geq 2x_2,\ x_4 \geq 2x_3\}$$

We introduce non-negative slack variables $y_j \in \mathbb{Z}_{\geq 0}$ to quantify the "excess" of each term relative to its lower bound.

$$x_1 = y_1 \tag{1.4}$$
$$x_2 = 2x_1 + y_2 = 2y_1 + y_2 \tag{1.5}$$
$$x_3 = 2x_2 + y_3 = 4y_1 + 2y_2 + y_3 \tag{1.6}$$
$$x_4 = 2x_3 + y_4 = 8y_1 + 4y_2 + 2y_3 + y_4 \tag{1.7}$$

The total weight is $S = x_1 + x_2 + x_3 + x_4 = 15y_1 + 7y_2 + 3y_3 + y_4$. Since y_j are independent and range freely over non-negative integers, the OGF $F(z)$ is the product of the individual generating functions for each y_j:

- For $15y_1$: $\sum_{y_1=0}^{\infty}(z^{15})^{y_1} = \frac{1}{1-z^{15}}$
- For $7y_2$: $\sum_{y_2=0}^{\infty}(z^{7})^{y_2} = \frac{1}{1-z^{7}}$
- For $3y_3$: $\sum_{y_3=0}^{\infty}(z^{3})^{y_3} = \frac{1}{1-z^{3}}$
- For y_4 : $\sum_{y_4=0}^{\infty}(z^{1})^{y_4} = \frac{1}{1-z^{1}}$

Multiplying these together yields the desired OGF:

$$F(z) = \frac{1}{(1-z)(1-z^3)(1-z^7)(1-z^{15})}$$

$$\{x_1 + x_2 \leq x_3\}$$

Following the same approach as in the previous problem, we use slack variables to convert the inequality into an equality of independent parts. The constraint $x_3 \geq x_1 + x_2$ can be rewritten by introducing a non-negative integer $y \geq 0$:

$$x_3 = x_1 + x_2 + y$$

The size of the composition is $S = x_1 + x_2 + x_3 + x_4 = 2x_1 + 2x_2 + y + x_4$. Since the variables x_1, x_2, y, x_4 are independent and each ranges freely over the non-negative integers, the OGF is the product of the generating functions for the individual terms in the sum:

- For $2x_1$: $\frac{1}{1-z^2}$
- For $2x_2$: $\frac{1}{1-z^2}$
- For y : $\frac{1}{1-z}$
- For x_4 : $\frac{1}{1-z}$

Multiplying these together gives:

$$F(z) = \frac{1}{(1-z^2)^2(1-z)^2}$$

$$\{x_1 + x_2 \geq x_3\}$$

Since the inequality does not immediately suggest a single slack variable, a more robust approach uses the complement method. Let $U(z) = (1-z)^{-4}$ be the OGF for four unrestricted variables. The universe of compositions partitions into those where $x_1+x_2 < x_3$ and our target class $x_1 + x_2 \geq x_3$. We find the OGF for the complement $x_1 + x_2 < x_3$. Set $x_3 = x_1 + x_2 + 1 + y$, where $y \geq 0$. The total sum S is:

$$S = x_1 + x_2 + (x_1 + x_2 + 1 + y) + x_4 = 2x_1 + 2x_2 + y + x_4 + 1$$

The OGF for this complement, which we denote by $C(z)$, is:

$$C(z) = \frac{z}{(1-z^2)^2(1-z)^2}$$

The OGF for our constraint is then:

$$F(z) = U(z) - C(z) = \frac{1}{(1-z)^4} - \frac{z}{(1-z^2)^2(1-z)^2} = \frac{1+z+z^2}{(1-z)^4(1+z)^2}$$

$$\{x_1 + x_2 \leq x_3 + x_4\}$$

To find the OGF for $\{x_1 + x_2 \leq x_3 + x_4\}$, we can use the complement method combined with the symmetry of the variables. Let $A = x_1 + x_2$ and $B = x_3 + x_4$. We seek the OGF for compositions satisfying $A \leq B$. In the universe of all possible compositions (x_1, x_2, x_3, x_4), there are three mutually exclusive cases: 1. $A < B$, 2. $A > B$, 3. $A = B$. By symmetry, the OGF for case 1 must be identical to the OGF for case 2, since A and B are built from the same number of variables with the same weights. Denote this common OGF by $G_<(z)$, and let $G_=(z)$ be the OGF for case 3. The total unrestricted OGF is $U(z) = \frac{1}{(1-z)^4}$. Therefore:

$$U(z) = 2G_<(z) + G_=(z)$$

Our target OGF is $F(z) = G_<(z) + G_=(z)$. Solving the system gives:

$$F(z) = \frac{U(z) + G_=(z)}{2}$$

Now $G_=(z)$ is the OGF for compositions in which the sum of the first two variables equals the sum of the last two.

- The number of compositions with $x_1 + x_2 = n$ is $[z^n]\frac{1}{(1-z)^2} = n + 1$.
- The number of compositions with $x_3 + x_4 = n$ is the same.
- For the equality $x_1 + x_2 = x_3 + x_4$ to hold, both sums must equal some $n \geq 0$.

$$G_=(z) = \sum_{n=0}^{\infty}(n+1)^2 z^{2n} = \frac{1+z^2}{(1-z^2)^3}$$

Substituting $U(z)$ and $G_=(z)$ into our formula yields:

$$F(z) = \frac{1}{2}\left[\frac{1}{(1-z)^4} + \frac{1+z^2}{(1-z^2)^3}\right] = \frac{1+z+2z^2}{(1-z)^4(1+z)^3}$$

$$\{x_1 \leq x_2,\ x_2 \geq x_3,\ x_3 \leq x_4\}$$

Because the inequalities alternate direction (up, down, up), we cannot use a straightforward chain of slack variable substitutions as we did for $x_1 \leq x_2 \leq x_3 \leq x_4$. Instead, we apply the inclusion–exclusion principle to the constraints. We begin with the universe of all unrestricted compositions, whose OGF is $U(z) = \frac{1}{(1-z)^4}$. We wish to enforce three conditions:

- $C_1 : x_1 \leq x_2$
- $C_2 : x_2 \geq x_3$
- $C_3 : x_3 \leq x_4$

However, it is simpler to work with chains of "less than or equal" inequalities. Consider the complement of the downward middle step. Take the set of all compositions satisfying both $x_1 \leq x_2$ and $x_3 \leq x_4$. From this set, we subtract those compositions that violate the middle constraint (i.e., $x_2 < x_3$).
The OGF for the set defined by $x_1 \leq x_2$ and $x_3 \leq x_4$: These two pairs are independent.

- For $x_1 \leq x_2$, set $x_2 = x_1 + y_1$. The total contribution is $2x_1 + y_1$, so the OGF is $\frac{1}{(1-z^2)(1-z)}$.
- For $x_3 \leq x_4$, the OGF is identically $\frac{1}{(1-z^2)(1-z)}$.

Since the two pairs are independent, we multiply their OGFs:

$$A(z) = \frac{1}{(1-z^2)^2(1-z)^2}$$

Now we find the OGF for the violation $x_1 \leq x_2 < x_3 \leq x_4$ by introducing slack variables:

- $x_1 = y_1$
- $x_2 = y_1 + y_2$
- $x_3 = y_1 + y_2 + 1 + y_3$
- $x_4 = y_1 + y_2 + 1 + y_3 + y_4$

Summing the four expressions gives:

$$S = 4y_1 + 3y_2 + 2y_3 + y_4 + 2$$

The OGF for this violation is therefore:

$$V(z) = \frac{z^2}{(1-z^4)(1-z^3)(1-z^2)(1-z)}$$

The OGF for the zigzag constraint is the OGF for the two independent pairs minus the OGF for the cases where the peak at x_2 fails:

$$\begin{aligned} F(z) &= \frac{1}{(1-z^2)^2(1-z)^2} - \frac{z^2}{(1-z^4)(1-z^3)(1-z^2)(1-z)} \\ &= \frac{1}{(1-z)(1-z^2)}\left[\frac{1}{(1-z)(1-z^2)} - \frac{z^2}{(1-z^3)(1-z^4)}\right] \end{aligned}$$

▷ **I.18.** *Odd versus distinct summands.* We establish a one-to-one correspondence between partitions into odd parts and partitions into distinct parts.

- Odd → Distinct
  1. Group identical odd parts together.
  2. If an odd part k appears m times, write m in binary: $m = 2^{a_1} + 2^{a_2} + \cdots$ with distinct powers of two.
  3. Replace the m copies of k by $2^{a_1} \cdot k,\ 2^{a_2} \cdot k,\ \ldots$
- Distinct → Odd
  1. Factor each part as $2^a \times$ odd.
  2. Split it into 2^a copies of that odd factor.
  3. Do this for all parts; now all parts are odd.
  4. Sort the resulting parts to obtain a partition into odd parts.

As an example:

$$\begin{aligned} 3+3+3+3+3+1+1 &= 5 \times 3 + 2 \times 1 \\ &= (2^2 + 2^0) \times 3 + 2^1 \times 1 \\ &= 4 \times 3 + 1 \times 3 + 2 \times 1 \\ &= 12 + 3 + 2 \end{aligned}$$

$$\begin{aligned} 12+3+2 &= 4 \times 3 + 3 + 2 \times 1 \\ &= (3+3+3+3) + 3 + (1+1) \\ &= 3+3+3+3+3+1+1 \end{aligned}$$

▷ **I.19.** *Euler's pentagonal number theorem.* To find a recurrence for the partition numbers, we consider the reciprocal of the generating function $P(z)$:

$$A(z) = \prod_{n=1}^{\infty}(1 - z^n)$$

Euler's Pentagonal Number Theorem states that this product expands into the following sparse power series:

$$\begin{aligned} A(z) &= \sum_{k=-\infty}^{\infty} (-1)^k z^{k(3k+1)/2} \\ &= \sum_{k=-\infty}^{\infty} (-1)^k z^{k(3k-1)/2} \\ &= 1 - z^1 - z^2 + z^5 + z^7 - z^{12} - z^{15} + \cdots \end{aligned}$$

The exponents $g_k = \frac{k(3k-1)}{2}$ are the generalized pentagonal numbers. The product of $P(z)$ and $A(z)$ must equal 1:

$$\sum_{n=0}^{\infty} p(n)z^n \cdot \sum_{k=-\infty}^{\infty} (-1)^k z^{g_k} = 1$$

Comparing the coefficient of z^n on both sides yields the algorithmically useful recurrence:

$$p(n) = p(n-1) + p(n-2) - p(n-5) - p(n-7) + p(n-12) + p(n-15) \cdots$$

A Python implementation is provided below:

```python
def compute_partitions(N):
P = [0] * (N + 1)
P[0] = 1  # Base case: p(0) = 1

for i in range(1, N + 1):
    k = 1
    while True:
        # Calculate pentagonal numbers for k and -k
        g_k_pos = k * (3 * k - 1) // 2
        g_k_neg = k * (3 * k + 1) // 2

        # Determine sign: +1 for k=1,2 (terms 1,2), -1 for k=3,4 (terms 3,4)...
        # Simplest way: (-1)^(k-1)
        sign = 1 if k % 2 == 1 else -1

        # Add p(i - g_k_pos)
        if g_k_pos <= i:
            P[i] += sign * P[i - g_k_pos]
        else:
            break # No more terms will fit

        # Add p(i - g_k_neg)
        if g_k_neg <= i:
            P[i] += sign * P[i - g_k_neg]
        else:
            break

        k += 1
return P
```

▷ **I.20.** *A digital surprise.*

$$\begin{aligned}
\varphi &= (1-10^{-1})(1-10^{-2})(1-10^{-3})(1-10^{-4})\cdots \\
&= \left[\prod_{n=1}^{\infty}(1-z^n)\right]_{z=.1} \\
&= \left[1 - z^1 - z^2 + z^5 + z^7 - z^{12} - z^{15} + z^{22} + z^{26} - \cdots\right]_{z=.1} \\
&= .\dot{9} - .1 - .01 + .00001 + .0000001 - .000000000001 - \cdots
\end{aligned}$$

The partial sums are:

- $\varphi_1 = .999999999999999999999999$
- $\varphi_2 = .899999999999999999999999$
- $\varphi_3 = .889999999999999999999999$
- $\varphi_4 = .890009999999999999999999$
- $\varphi_5 = .890010099999999999999999$

- $\varphi_6 = .89001009999899999999999$
- $\varphi_7 = .89001009999899899999999$
- $\varphi_8 = .89001009999899990000000999$

▷ **I.21.** *Lattice points.* We are interested in the number of integer solutions to the inequality $\sum_{i=1}^{d} x_i^2 \leq n^2$. First, consider a single coordinate $x_i \in \mathbb{Z}$. The "size" contributed by this coordinate to the sum of squares is x_i^2. The generating function for the possible values of a single coordinate, where the exponent of z represents the square of the coordinate, is:

$$\Theta(z) = \sum_{x \in \mathbb{Z}} z^{x^2} = 1 + 2\sum_{n=1}^{\infty} z^{n^2}$$

Since each coordinate $x_1, \cdot, x_d$ is chosen independently, the generating function for the sum of squares $\sum x_i^2$ is the d-th power of the individual generating function:

$$S(z) = (\Theta(z))^d = \sum_{k=0}^{\infty} r_d(k) z^k$$

In this series, the coefficient $[z^k]S(z)$ (denoted as $r_d(k)$) is exactly the number of ways to represent the integer k as a sum of d squares. Geometrically, this is the number of lattice points on the sphere (the boundary) of radius $\sqrt{k}$. The number of points $N(n, d)$ in the closed ball of radius n is:

$$N(n, d) = \sum_{k=0}^{n^2} r_d(k)$$

In generating functions, the prefix sum of a sequence's coefficients is obtained by multiplying the generating function by $\frac{1}{1-z}$. Therefore:

$$\sum_{k=0}^{\infty} \left(\sum_{j=0}^{k} r_d(j) \right) z^k = \frac{1}{1-z} S(z) = \frac{1}{1-z} (\Theta(z))^d$$

To find the total count up to $k = n^2$, we simply extract the coefficient of z^{n^2}:

$$N(n, d) = [z^{n^2}] \frac{1}{1-z} (\Theta(z))^d$$

1.4 Words and regular languages

▷ **I.23.** *Alice, Bob, and coding bounds.* Following Proposition I.4, the class $\mathcal{C}$ of binary strings avoiding the pattern 11 has the OGF:

$$C(z) = \frac{1+z}{1-z-z^2}$$

To account for a variable-length coding scheme where codewords are selected from $\bigcup_{j=0}^{L} \mathcal{C}_j$, we consider the sum of the first $L+1$ coefficients. This is represented by the OGF:

$$S(z) = \frac{C(z)}{1-z} = \frac{1+z}{(1-z)(1-z-z^2)}$$

The asymptotic growth of the sum $\sum_{j=0}^{L} C_j$ is governed by the smallest singularity of $S(z)$. Since the pole at $z = 1/\varphi$ (where $\varphi = \frac{1+\sqrt{5}}{2}$) is closer to the origin than the pole at $z = 1$, the exponential growth is dominated by $C(z)$. Specifically, as $L \to \infty$:

$$\sum_{j=0}^{L} C_j \sim A \cdot \varphi^L$$

for some constant A. To find the required code length $L(n)$ for n bits of information, we solve for $2^n \leq \sum_{j=0}^{L} C_j$ by taking the base-2 logarithm:

$$n \leq \log_2 \left(\sum_{j=0}^{L} C_j \right) = L \log_2 \varphi + O(1)$$

Rearranging for L, we obtain the lower bound:

$$L \geq \frac{n}{\log_2 \varphi} + O(1) \approx 1.440420n + O(1)$$

▷ **I.27.** *Congruence languages.* The DFA $\mathcal{A}$ for binary strings congruent to 2 (mod 7) is structured as follows:

- **States**: A set $\mathcal{Q} = \{S_0, \ldots, S_6\}$, where each state S_r represents the current remainder r modulo 7.
- **Initial State**: S_0 (representing the value 0).
- **Accepting State**: $\overline{Q} = \{S_2\}$.
- **Alphabet**: $\mathcal{A} = \{0, 1\}$.

Each transition follows the rule $q_r \circ b = q_{(2r+b) \pmod 7}$:

Table 1.1: State Transition

Current State (Remainder)	Input 0 ($2r \pmod 7$)	Input 1 ($2r + 1 \pmod 7$)
S_0	S_0	S_1
S_1	S_2	S_3
S_2	S_4	S_5
S_3	S_6	S_0
S_4	S_1	S_2
S_5	S_3	S_4
S_6	S_5	S_6

▷ **I.29.** *The forward equations.* By a "last-letter decomposition", the set $\mathcal{M}_k$ of words leading to state q_k from the initial state q_0 satisfies the combinatorial relation:

$$\mathcal{M}_k \cong [\![k = 0]\!] \cdot \{\epsilon\} + \sum_{j,\alpha : q_j \circ \alpha = q_k} (\mathcal{M}_j \times \{\alpha\})$$

where $[\![\cdot]\!]$ is the Iverson bracket. This translates to a linear system of OGFs:

$$M_k(z) = \delta_{k,0} + z \sum_j M_j(z) T_{j,k}$$

In vector notation, with $\mathbf{M}(z)$ as a column vector and $\mathbf{e}_0$ the first basis vector, the system is a transposed version of the backward equations:

$$\mathbf{M}(z)^T = \mathbf{e}_0^T + z\mathbf{M}(z)^T T$$

▷ **I.33.** *Waiting times in strings.* To illustrate the formulas, let's define a prefix language $\hat{\mathcal{L}}$ containing one word of length 2 and one word of length 3. Let $\hat{\mathcal{L}} = \{aa, bbb\}$.

1. Verification of Prefix Property: The word aa is not a prefix of bbb, and bbb is not a prefix of aa. Since no word is a strict prefix of another, this is a valid prefix language $\hat{\mathcal{L}}$.

2. The Generating Function $\hat{L}(z)$: The OGF is the sum of z raised to the power of each word's length: $\hat{L}(z) = z^2 + z^3$.

3. Probability: $\hat{L}(1/2)$: This represents the probability that a random infinite string starts with a word in $\mathcal{L}$: $\hat{L}(1/2) = (1/2)^2 + (1/2)^3 = 1/4 + 1/8 = 3/8$. This means there is a 37.5% chance that a random string begins with a word from this language.

4. Expected Time: $\frac{1}{2}\hat{L}'(1/2)$. To find the expected time (or more precisely, the expected "matched length") at which a word in $\mathcal{L}$ is first encountered: $\hat{L}'(z) = 2z + 3z^2 \implies \hat{L}'(1/2) = 1.75 \implies E = 0.875$.

5. Breakdown of the Result: The total expected value of 0.875 is formed as follows:

   - Word "aa": Probability 1/4 of length 2 → Contribution: $2 \times 0.25 = 0.5$.
   - Word "bbb": Probability 1/8 of length 3 → Contribution: $3 \times 0.125 = 0.375$.
   - No Match: Probability 5/8 of length 0 → Contribution: $0 \times 0.625 = 0$.

▷ **I.34.** *A probabilistic paradox on strings.* Consider a binary pattern of length k. The generating function $T(z)$ for the first occurrence of this pattern is given by:

$$T(z) = \frac{z^k}{z^k + (1 - mz)c(z)} = \frac{z^k}{z^k + (1 - 2z)c(z)}$$

The average waiting time (or expected occurrence time) can be derived directly from $T(z)$ by evaluating its derivative at the point of the alphabet's inverse size ($z = 1/2$):

$$\left[z \frac{d}{dz} T(z) \right]_{z=1/2} = 2^k c(1/2)$$

1.5 Tree structures

▷ **I.39.** *Unary–binary trees and Motzkin numbers.* We identify the characteristic polynomial as $\phi(u) = 1 + u + u^2$. To find the number of trees with n nodes, $M_n = [z^n]M(z)$, we apply the Lagrange Inversion Theorem:

$$\begin{aligned}
M_n &= \frac{1}{n}[u^{n-1}]\phi(u)^n \\
&= \frac{1}{n}[u^{n-1}](1+u+u^2)^n \\
&= \frac{1}{n}[u^{n-1}]\sum_j \binom{n}{j}(u+u^2)^j \\
&= \frac{1}{n}\sum_j \binom{n}{j}[u^{n-1}]u^j(1+u)^j \\
&= \frac{1}{n}\sum_j \binom{n}{j}[u^{n-j-1}]\sum_l \binom{j}{l}u^l \\
&= \frac{1}{n}\sum_j \binom{n}{j}\binom{j}{n-j-1} \\
&= \frac{1}{n}\sum_k \binom{n}{n-k}\binom{n-k}{n-(n-k)-1} \\
&= \frac{1}{n}\sum_k \binom{n}{k}\binom{n-k}{k-1}
\end{aligned}$$

▷ **I.42.** *Faster determination of Schröder numbers.*

$$S(z) = \frac{1}{4}(1 + z - \sqrt{1 - 6z + z^2})$$

Consider in general

$$p(z)S(z) + q(z) = \sqrt[k]{r(z)}$$

where $p(z)$, $q(z)$, and $r(z)$ are functions of z. Taking the derivative with respect to z,

$$p'(z)S(z) + p(z)S'(z) + q'(z) = \frac{1}{k}r(z)^{\frac{1}{k}-1}r'(z)$$

Multiplying both sides by $r(z)$,

$$p'(z)r(z)S(z) + p(z)r(z)S'(z) + q'(z)r(z) = \frac{1}{k}(p(z)S(z) + q(z))r'(z)$$

Rearranging,

$$p(z)r(z)S'(z) + (p'(z)r(z) - \frac{1}{k}p(z)r'(z))S(z) = \frac{1}{k}q(z)r'(z) - q'(z)r(z)$$

Plugging $p(z) = -4$, $q(z) = 1 + z$, $k = 2$, and $r(z) = 1 - 6z + z^2$ into the equation and simplifying yields

$$(z^2 - 6z + 1)S'(z) + (3 - z)S(z) = 1 - z$$

from which the recurrence follows.

1.6 Additional constructions

1.7 Perspective

Chapter 2

LABELLED STRUCTURES AND EXPONENTIAL GENERATING FUNCTIONS

2.1 Labelled classes

2.2 Admissible labelled constructions

2.3 Surjections, set partitions, and words

2.4 Alignments, permutations, and related structures

2.5 Labelled trees, mappings, and graphs

2.6 Additional constructions

2.7 Perspective

Chapter 3

COMBINATORIAL PARAMETERS AND MULTIVARIATE GENERATING FUNCTIONS

Content for chapter 3 goes here...

3.1 An introduction to bivariate generating functions (BGFs)

3.2 Bivariate generating functions and probability distributions

3.3 Inherited parameters and ordinary MGFs

3.4 Inherited parameters and exponential MGFs

3.5 Recursive parameters

3.6 Complete generating functions and discrete models

3.7 Additional constructions

3.8 Extremal parameters

3.9 Perspective